

Which statement is true regarding the rank and null space of a matrix A ?

Choose the correct option

- ☐ The rank of A is equal to the dimension of its null space.
- ☒ The null space of A contains all the vectors that are orthogonal to the row space of A .
- ☐ The rank of A plus the dimension of its null space is always equal to the number of rows of A .
- ☐ A matrix with full rank has a non-trivial null space.

Clear

Next

Problem 4 (Convex Optimization - Convex Function). Which of the following functions is convex over its entire domain? Assume x and y are real numbers.

Choose the correct option

- ☐ $f(x, y) = x^2 + y^2 - xy$
- ☐ $f(x, y) = x^2 - y^2$
- ☒ $f(x, y) = e^x + e^y$
- ☐ $f(x, y) = \log(x^2 + y^2)$

Clear

Next

Question: 24 of 24

In the context of ensemble learning, particularly with Random Forests, which statement best describes how the ensemble method affects the bias-variance tradeoff?

Choose the correct option

- ☐ Increasing the number of trees in a Random Forest will primarily reduce the bias of the model, as each tree contributes to a higher bias reduction.
- ☐ As the number of trees in a Random Forest increases, the variance of the ensemble model will decrease, but the bias will increase significantly.
- ☒ The Random Forest algorithm mitigates high variance by averaging predictions across multiple trees, which reduces the variance while maintaining a similar level of bias compared to a single decision tree.
- ☐ Adding more features to the Random Forest model always reduces both bias and variance simultaneously, leading to better overall model performance.

[Clear](#)

This is the last question in this section. Please click Finish Test.

Problem 10 (Generative Adversarial Networks (GANs)). **True or False:** In a GAN with a discriminator D and a generator G , if the discriminator is achieving perfect classification (i.e., $D(x) = 1$ for all real data x , and $D(G(z)) = 0$ for all generated data $G(z)$), the gradient of the generator's loss with respect to its parameters will be zero, assuming the standard GAN loss functions are used.

Given:

- The discriminator's output $D(x)$ represents the probability that x is real data
- The generator's loss function is: $L_G = \mathbb{E}_z[\log(1 - D(G(z)))]$
- The discriminator's loss function is: $L_D = -\mathbb{E}_x[\log(D(x))] - \mathbb{E}_z[\log(1 - D(G(z)))]$
- z is the random noise input to the generator

Choose the correct option

- ☒ True
- ☐ False

Clear

Next

Training neural networks implicitly perform representation learning. However, this representation of learning can neither be categorized as supervised learning nor as unsupervised learning.

Choose the correct option

☐ True

☒ False

Clear

Next

Question: 15 of 24

Problem 3 (Neural Networks and Backpropagation). True or False: In a neural network using the softmax activation function for multiclass classification, the gradient of the cross-entropy loss with respect to the pre-activation outputs is always equal to the difference between the predicted probabilities and the true labels, regardless of the number of classes.

Choose the correct option.

- ☒ True
☐ False

Clear

Next

Question: 18 of 24

The binary classification task is under hinge-loss. Let F_1 be the l_2 regularized linear model with a margin of at least 1 and let F_2 be the l_2 regularized linear model with a margin of at least 2. Then, the standard estimation error bound with F_1 is $\leq$ that with F_2 .

Choose the correct option

☐ True☒ False

Clear

Next

Which option correctly explains a key aspect of the loss function used in Variational Autoencoders (VAEs)?

Choose the correct option

- ☐ The VAE loss function combines a reconstruction loss with a cross-entropy loss to encourage accurate data reconstruction.
- ☐ The VAE loss function consists of a KL divergence term that encourages the latent variables to approximate a sparse representation of the data.
- ☐ The reconstruction term in the VAE loss function ensures that the latent space is regularized to follow a uniform distribution.
- ☒ The VAE loss function includes a KL divergence term that forces the approximate posterior distribution of the latent variables to be close to a prior distribution, usually a standard normal distribution.

Clear

Next

Choose the correct option

In a Support Vector Machine (SVM) with a soft margin, which of the following equations represents the objective function to be minimized?

- ☒ $\frac{1}{2}\|w\|^2 + C \sum_{i=1}^n \xi_i$
- ☐ $\frac{1}{2}\|w\|^2 + C \sum_{i=1}^n \xi_i^2$
- ☐ $\frac{1}{2}\|w\|^2 - C \sum_{i=1}^n \xi_i$
- ☐ $\frac{1}{2}\|w\|^2 + \sum_{i=1}^n \xi_i$

Clear

Next

Problem 6 (Statistics (Law of Large Numbers)). Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$. Define $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Which of the following best describes the convergence of $\bar{Y}_n$ to μ as n approaches infinity?

Choose the correct option

- ☐ $\bar{Y}_n$ converges to μ in probability
- ☐ $\bar{Y}_n$ converges to μ almost surely
- ☐ $\bar{Y}_n$ converges to μ in distribution
- ☒ All of the Above

Clear

Next

Question: 22 of 24

In tokenization strategies for large language models, which statement accurately distinguishes the WordPiece and Byte Pair Encoding (BPE) methods?

Choose the correct option

- ☒ WordPiece and BPE both operate by iteratively merging the most frequent pairs of characters or subwords, but WordPiece also includes a mechanism to handle out-of-vocabulary words by splitting them into known subword units.
- ☐ Byte Pair Encoding (BPE) starts with individual characters and merges the most frequent pairs of characters to form subword units, whereas WordPiece begins with whole words and splits them into character-level units if they are not present in the vocabulary.
- ☐ WordPiece uses a vocabulary built from a mixture of character-level and subword units, where the subword units are derived from the most frequent pairs of characters, while BPE exclusively uses subword units derived from merging frequent pairs without considering characters.
- ☐ Byte Pair Encoding (BPE) and WordPiece both use a pre-defined vocabulary of subword units and apply a fixed set of rules for tokenization; however, WordPiece utilizes a probabilistic model to choose between subword segments based on their likelihood of occurring in the corpus.

Clear

Next

Finish Test

Question: 23 of 24

The binary classification task is under hinge-loss. Let F_1 be the l_2 regularized linear model with a margin of at least 1 and let F_2 be the l_2 regularized linear model with a margin of at least 2. Then, the modeling error with F_1 is $\leq$ that with F_2 .

Choose the correct option

- ☐ True
- ☒ False

Clear

Next

In Large Language Models (LLMs) that utilize autoregressive modeling, which statement is most accurate?

Choose the correct option

- ☐ Autoregressive models generate the entire sequence at once, rather than token by token.
- ☐ In autoregressive models, the probability of generating the next token depends solely on the last token generated.
- ☒ Autoregressive models can generate highly coherent text by conditioning the generation of each token on the previously generated tokens.
- ☐ Autoregressive models are typically trained using a bidirectional context to predict the next word in a sequence.

Clear

Next

Question: 13 of 24



Choose the correct option

Problem 5 (Random walks). In a symmetric random walk on a line, starting from position 0, what is the probability of returning to the starting point after $2n$ steps?

☒ $\frac{\binom{2n}{n}}{2^{2n}}$

☐ $\frac{\binom{2n}{n}}{2^n}$

☐ $\frac{n}{2^{2n}}$

☐ $\frac{1}{2^n}$

Clear

Next

In evaluating a binary classification model, you are given the following precision and recall metrics: Precision: 0.75, Recall: 0.60. If you were to increase the decision threshold for classifying an instance as positive, which statement best describes the expected impact on precision and recall?

Choose the correct option

- ☐ Increasing the decision threshold will increase both precision and recall, as the model becomes more confident in its positive predictions.
- ☐ Increasing the decision threshold will decrease both precision and recall, as the model will classify fewer instances as positive, leading to more false negatives and potentially fewer true positives.
- ☒ Increasing the decision threshold will increase precision but decrease recall, as the model becomes more conservative about classifying instances as positive, reducing false positives but also potentially missing true positives.
- ☐ Increasing the decision threshold will decrease precision but increase recall, because the model will classify more instances as positive, capturing more true positives but also increasing false positives.

Clear

Next

Question: 7 of 24

Let $k_1: X \times X \rightarrow \mathbb{R}$ and $k_2: X \times X \rightarrow \mathbb{R}$ be two valid kernels over the domain, X . Define a new function k by: $k(z) = k_1(z)k_2(z)$ for all z belongs to $X \times X$. Then, k is a valid kernel over X .

Choose the correct option

☒ True

☐ False

Clear

Next

Typical estimation error bounds for regularized linear models are independent of input-dimensionality.

Choose the correct option

☐ True

☒ False

Clear

Next

Which statement is true about using the Radial Basis Function (RBF) kernel in a Support Vector Machine (SVM) model?

Choose the correct option

- ☐ The RBF kernel is effective only when the data is linearly separable.
- ☒ The RBF kernel introduces a non-linear transformation to map the input data into a higher-dimensional space.
- ☐ The RBF kernel is less computationally intensive than the linear kernel.
- ☐ The RBF kernel cannot be used with datasets that have more than two classes.

Clear

Next

Finish Test

Choose the correct option

Which of the following statements is true about the rank of a matrix A and its SVD components?

- ☒ The rank of A is equal to the number of non-zero singular values in Σ .
- ☐ The rank of A is the number of non-zero entries in U .
- ☐ The rank of A is equal to the number of non-zero entries in V .
- ☐ The rank of A is the sum of the number of rows and columns of Σ .

Clear

Next

Which of the following statements is true about the rank of a matrix A and its SVD components?

Choose the correct option

- ☒ The rank of A is equal to the number of non-zero singular values in Σ .
- ☐ The rank of A is the number of non-zero entries in U .
- ☐ The rank of A is equal to the number of non-zero entries in V .
- ☐ The rank of A is the sum of the number of rows and columns of Σ .

Clear Next

In the context of encoder-decoder models used for sequence-to-sequence tasks in NLP, which statement accurately describes the role and impact of the attention mechanism on the model's performance?

Choose the correct option

- ☐ The attention mechanism replaces the need for an encoder and decoder, directly mapping input sequences to output sequences with a fixed-size representation.
- ☒ The attention mechanism allows the decoder to focus on different parts of the input sequence dynamically, improving performance by mitigating the bottleneck of fixed-size context vectors in traditional encoder-decoder architectures.
- ☐ The attention mechanism reduces the model's computational complexity by limiting the context window to a fixed number of tokens in the input sequence, thus simplifying the encoding process.
- ☐ The attention mechanism primarily enhances the encoder's ability to capture long-range dependencies by providing a global context that is fixed throughout the entire decoding process.

Clear

Next

Problem 8 (Attention in Neural Networks). In the context of self-attention mechanism, given a query vector q , a key vector k , and a value vector v , which of the following expressions correctly represents the attention score before softmax normalization?

Choose the correct option.

☒ $q^T \cdot k$

☐ $q \cdot k^T$

☐ $q^T \cdot v$

☐ $k^T \cdot v$

Clear

Next

Linear models are not universal approximators, whereas kernel-based models are universal approximators.

Choose the correct option

- ☒ True
☐ False

Clear

Next